

PIYUSH GUPTA

Full Stack Developer

piyushgbra@gmail.com | 9975195044 | India | [GitHub](#) | [LinkedIn](#)

SUMMARY

Computer Science undergraduate focused on full-stack engineering, with hands-on experience designing and building responsive, scalable web applications end-to-end. Comfortable across the stack — JavaScript/TypeScript, React, Node.js, Express, MongoDB/SQL, and REST API design — with a strong grounding in data structures, algorithms, and system fundamentals. Enjoys writing clean, well-tested code and is actively deepening knowledge of system design and cloud infrastructure.

TECHNICAL SKILLS

Languages: C, C++, JavaScript, TypeScript, Python, SQL

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: MongoDB, MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Docker, Postman, VS Code

Foundations: OOP, DBMS, Data Structures & Algorithms, Operating Systems, Computer Networks

PROJECTS

Decentralized Notary App

React, Node.js, Express, MongoDB

- Built a full-stack document verification platform enabling users to upload, hash, and validate document authenticity.
- Implemented SHA-256 hashing to guarantee document integrity and detect tampering without storing raw file contents.
- Designed and exposed REST APIs for upload, verification, and record management, following clean resource-based conventions.
- Built a responsive, accessible UI so verification works consistently across desktop and mobile.

Eye-Blink Controlled HCI for Augmentative & Alternative Communication

Arduino (C++), Python, HTML5/CSS3/JavaScript, Web Speech API

- Designed an assistive communication system enabling users with severe motor disabilities to communicate using only eye blinks.
- Built an Arduino-based sensing module using dual IR proximity sensors to reliably detect and debounce left, right, and both-eye blinks in real time.
- Developed a Python serial bridge (pySerial, keyboard/pynput) to translate blink signals from the microcontroller into keyboard events.
- Built a tab-based, fully blink-navigable web interface with scanning-style selection and integrated the Web Speech API for text-to-speech output.

Digital Voting System for Student Council Selection

MySQL, Python (Tkinter)

- Built a desktop voting application with a Tkinter GUI, covering candidate registration, voting, and result views.
- Modeled a relational MySQL schema to reliably store voter, candidate, and result data.
- Implemented duplicate-vote prevention and real-time tallying so results stay accurate as votes are cast.

CERTIFICATIONS & CONTINUOUS LEARNING

Python | Saylor Academy | [Certificate](#)

Soft Skills | Alison | [Certificate](#)

MERN Stack Development — currently in progress

EDUCATION

B.Tech, Computer Science & Engineering (Hons.) — Full Stack Development

2025 – 2029

Lovely Professional University, in collaboration with GeeksforGeeks

Dhruv Global School

2023 – 2025